CSE302: Database Systems (Section 02) [SUMMER 2022]

Project Report

Project/Application Title: Capstone Project

Project URL/Link: https://apex.oracle.com/pls/apex/r/akjk/cse302_capstone_project

Submitted by:

Student ID	Student Name
2020-1-60-053	KAZI FERDOUS HASAN
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1. Final version of the E-R Model of the Project

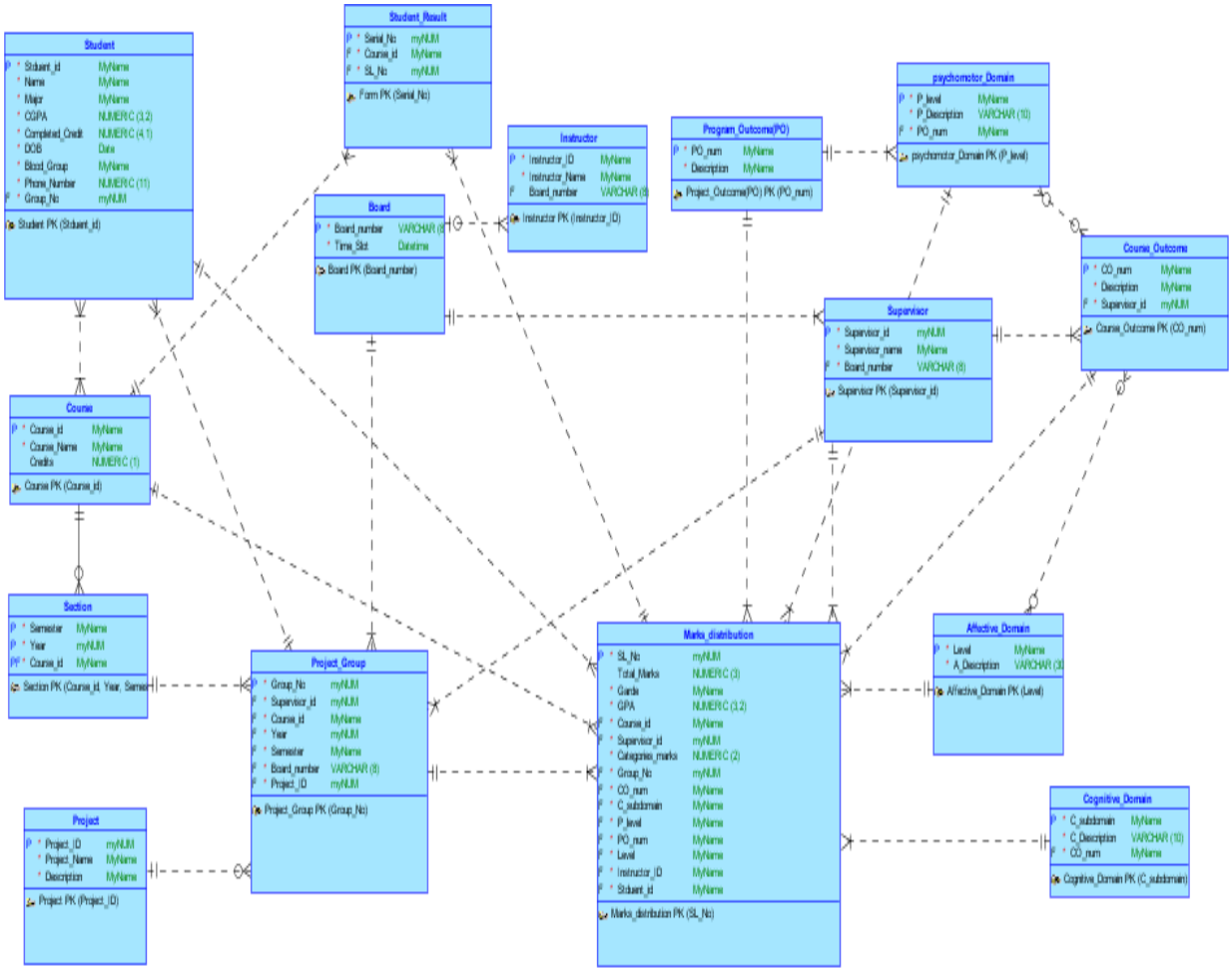

Figure:01

We have assumed that one project can be assigned in different project groups or not and every project group must have at least 1 project id.
We have assumed that one student must assigned in
one project group and one project group may have multiple students or maybe one student.
We have assumed that in marks_distribution one student may have multiple marks for different criteria and one mark must be assigned to at least one student.
We have assumed that one student can take many courses in one semester and also one course can be assigned for every student so that the relation between student and course is many to many relation.
One affective domain under in many course outcomes (CO) or not and similarly one course outcome may have many affective domains or not.
One psychomotor domain under many course outcomes (CO) or not and also one course outcome has many psychomotor domains or not.

In the marks distribution table, we assumed that one supervisor can assign marks for each and every student and one mark is given by one supervisor.
Similarly, one instructor can give many marks to every student and one mark is given by at least one instructor.

2. Final Version of the Relational Data Model of the Project

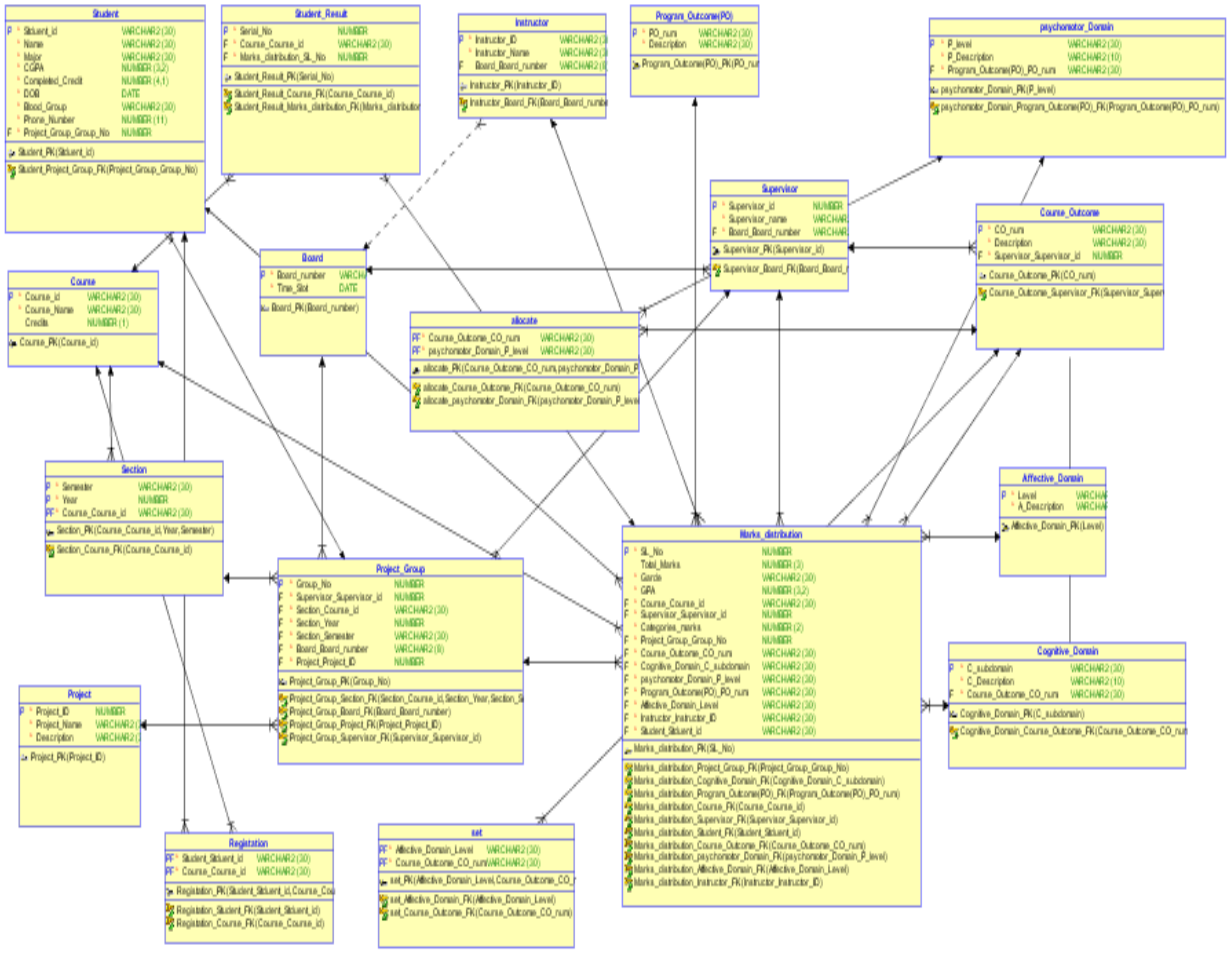

Figure:02

We get a registration table by applying many to many conditions on the student and course table. The registration table keeps track of particular students taking particular courses.
From our assumption, we get a set table by establishing many to many relations in between course outcome and affective domain table which contains primary key both of the table.
We get an allocated table by establishing many to many relations in between course outcome and psychomotor domain table which contains the primary key of both of the tables.

3. Implementation Checklist

Description	**Remarks**
Project ID	Application 152569 CSE302_Capstone_project
Workspace Name	AKJK
Workspace Email	akjk10101010@gmail.com
Workspace Password	AKjk01010101;
Authentication Type	Application Express/Custom (keep the relevant one)
Username and Password to Login	admin and admin123
write username like admin	student and student123
if you have multiple users, write all username and password correctly	instructor and instructor123
Uploaded and Displayed Images in Reports	yes
Implemented Page-wise Authorization	yes
Number of Reports with Forms	31
Number of Master-detail Forms	3
Number of Reports based on Queries, Aggregate Queries	
Number of Parameterized Reports	
Number of Graphs/Charts	
Included any trigger/other feature which is not taught in class. If yes, please explain briefly.	
Write the most unique feature/functionality in your application or the page of your application that you are proud of. Explain briefly.	

4. Reports with Forms

Report Page Number and Name	Form Page Number and Name	Table
45 and student report	40 and student form	student
47 and section report	14 and section form	section
49 and project report	35 and project form	project
55 and instructor report	103 and instructor form	instructor
5 and course report	11 and course form	course
61 and supervisor report	18 and supervisor form	supervisor
53 and course_outcome report	22 and course_outcome form	course_outcome
57 and program_outcome report	27 and program_outcome form	program_outcome
78 and supervisor_image_upload report	77 and supervisor_image_form	supervisor_image
70 and instructor_image_upload report	69 and instructor_image_form	Instructor_image
126 and marks distribution report	38 and marks_distribution form	Marks_distribution

student_Report

Stdent ID	Name	Major	Cgpa	Completed Credit	Dob	Blood Group	Phone Number	Project Group Group No
2018-2-60-100	Sabbir Ahmed	Data Science	3.52	105	8/15/1999	B+	1736989734	123
2018-2-60-106	Md. Abdul Kader Jilani	Data Science	3.51	109	1/12/1999	AB-	1786613879	123
2018-2-60-011	Sadia Afrin	Data Science	3.52	109	2/20/1998	A+	1967875006	121
2018-3-60-031	Tahmid Alam	Data Science	3.59	110	6/10/1999	O+	1623227527	122
2018-2-60-077	Md. Salem Talukder	Data Science	3.56	105	8/10/1999	O+	1686557797	122
2018-2-60-110	Mohammad Khalid Hasan	Data Science	3.58	111	6/11/1999	AB+	1799041090	123
2018-2-60-013	Lata Rani Saha	Data Science	3.51	110	6/20/1998	A+	1730890352	121
2018-2-60-008	Shamima Yesmin	Data Science	3.66	105.5	1/20/1998	A+	1628325530	121
2018-2-60-021	Shahriar Chowdhury	Data Science	3.7	107	7/20/1999	AB-	1681689067	121
2018-2-60-097	Moumita Das	Data Science	3.58	105	7/5/1999	A+	1521351018	122
2018-2-60-101	Md. Nayem Ahmed	Data Science	3.5	101	1/5/1999	B-	1521105007	123
2018-2-60-047	Arnob Ghosh	Data Science	3.58	111	8/15/1999	O-	1925757002	122

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Figure:4.1

Section_Report

Semester	Course Course	Year
Fall20	Capstone Project I	2020
Spring20	Capstone Project I	2020
Summer20	Capstone Project I	2020
Fall21	Capstone Project I	2021
Spring21	Capstone Project I	2021
Summer21	Capstone Project I	2021
Fall20	Capstone Project II	2020
Spring20	Capstone Project II	2020
Summer20	Capstone Project II	2020
Fall21	Capstone Project II	2021
Spring21	Capstone Project II	2021
Summer21	Capstone Project II	2021
Fall20	Capstone Project III	2020
Spring20	Capstone Project III	2020
Summer20	Capstone Project III	2020
Fall21	Capstone Project III	2021
Spring21	Capstone Project III	2021
Summer21	Capstone Project III	2021

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Figure:4.2

project_Report

Go Actions

Project ID	Project Name	Description
33	Detection of Road Lane Lines	Using built-in Python library
11	Fake News Detection	Using Python & TfidfVectorizer
22	Detecting Forest Fire	Using k-means clustering

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Figure:4.3

Instructor_Report

Go Actions

Instructor ID	Instructor Name	Board Board Number
1110	Dr. Shamim H Ripon	Board1
1100	Dr.Md. Nawab Yousuf Ali	Board3
1000	Dr. Ahmed Wasif Reza	Board2
2000	Dr. Md Sawkat Ali	Board4
1111	Dr.Md. Mozammel Huq Azad Khan	Board1
2200	Dr. Maheen Islam	Board2
2220	Dr. Taskeed Jabid	Board4
2222	Dr. Mohammad Rezwanul Huq	Board3

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Figure:4.4

Supervisor_Report

Go Actions

Supervisor ID	Supervisor Name	Board Board Number
2222	Dr. Mohammad Rezwanul Huq	Board3
2220	Dr. Taskeed Jabid	Board4
2200	Dr. Maheen Islam	Board2
1111	Dr.Md. Mozammel Huq Azad Khan	Board1

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Figure:4.5

COURSE_OUTCOME REPORT

Co Num	Description	Supervisor Supervisor
CO2	Examine, define & formulate	Dr. Mohammad Rezwanul Huq
CO4	Design and develop solutions	Dr. Mohammad Rezwanul Huq
CO7	Assess & address sustainable	Dr.Md. Mozammel Huq Azad Khan
CO10	Write effective report	Dr.Md. Mozammel Huq Azad Khan
CO12	Prepare for life-long learning	Dr.Md. Mozammel Huq Azad Khan
CO1	Integrate acquired knowledge	Dr. Mohammad Rezwanul Huq
CO3	Analyze various solution	Dr. Mohammad Rezwanul Huq
CO5	Identify & apply modern tools	Dr. Mohammad Rezwanul Huq
CO6	Asses & address all aspects	Dr. Mohammad Rezwanul Huq
CO8	Apply professional principles	Dr.Md. Mozammel Huq Azad Khan
CO11	Conduct cost estimation;	Dr.Md. Mozammel Huq Azad Khan
CO9	Work in team & individually	Dr.Md. Mozammel Huq Azad Khan

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Figure:4.6

PROGRAM_OUTCOME REPORT

Po Num	Description
PO1	Engineering Knowledge
PO2	Problem Analysis
PO3	Development of Solutions
PO5	Modern Tool Usage
PO6	The Engineer and Society
PO7	Environment and Sustainability
PO8	Ethics
PO10	Communication
PO11	Project Management and Finance
PO4	Investigation
PO9	Individual Work and Teamwork
PO12	Life-Long Learning

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App 152569 Page 57 Session Debug Quick Edit Customize

Figure:4.7

supervisor image_report

Supervisor ID	Image
2200	
1111	

Figure:4.8

supervisor image_report

Supervisor ID	Image
2222	
2220	

Figure:4.9

Insturctor_image_display

Instructor ID	Image
1111	
2222	

Figure:4.10

Insturctor_image_display

Instructor ID	Image
1000	
2000	

Figure:4.11

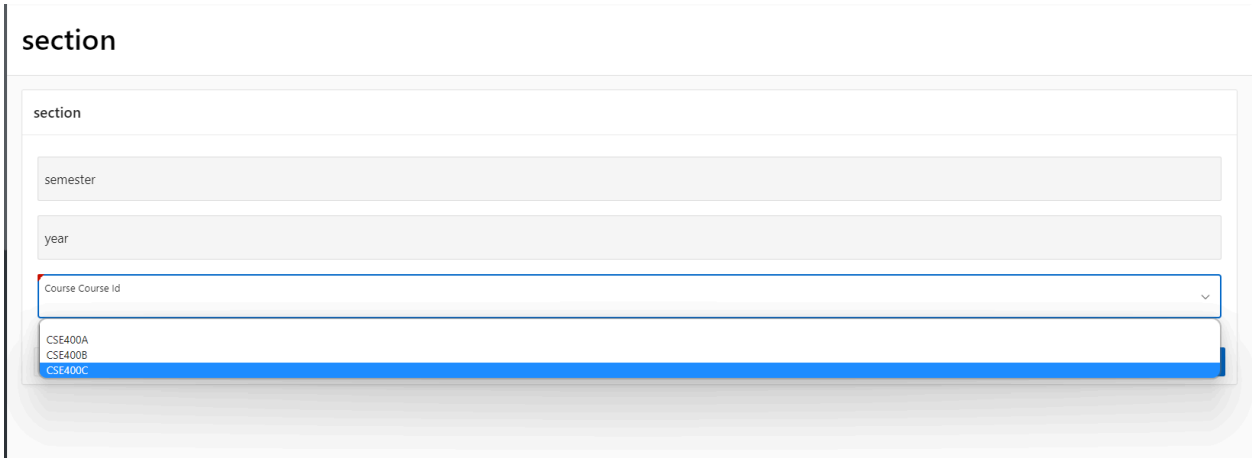
Insturctor_image_display

Instructor ID	Image
2200	
2220	

Figure:4.12

STUDENT_FORM

Figure:4.13



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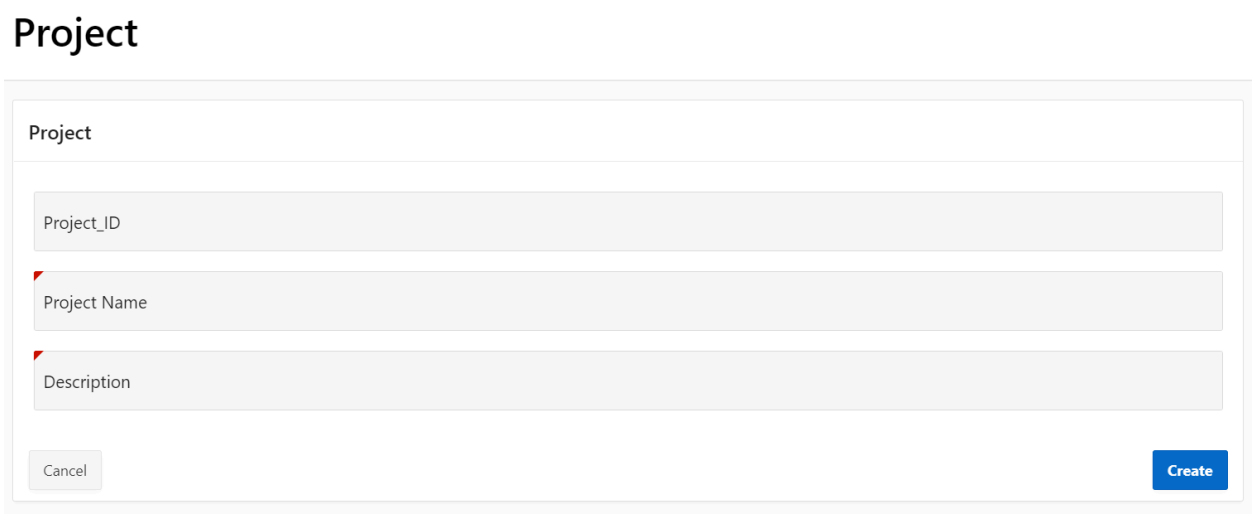

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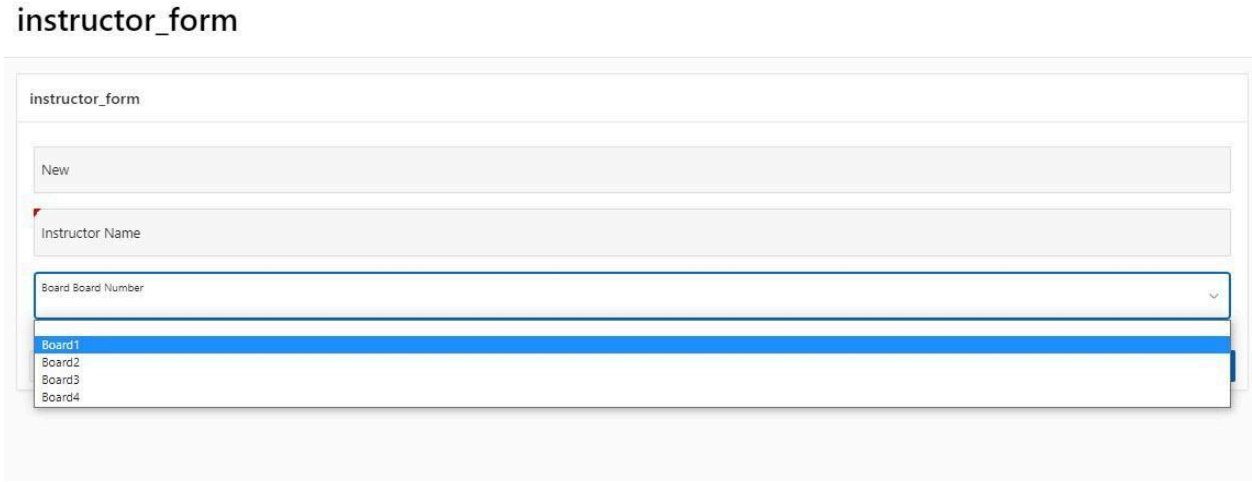

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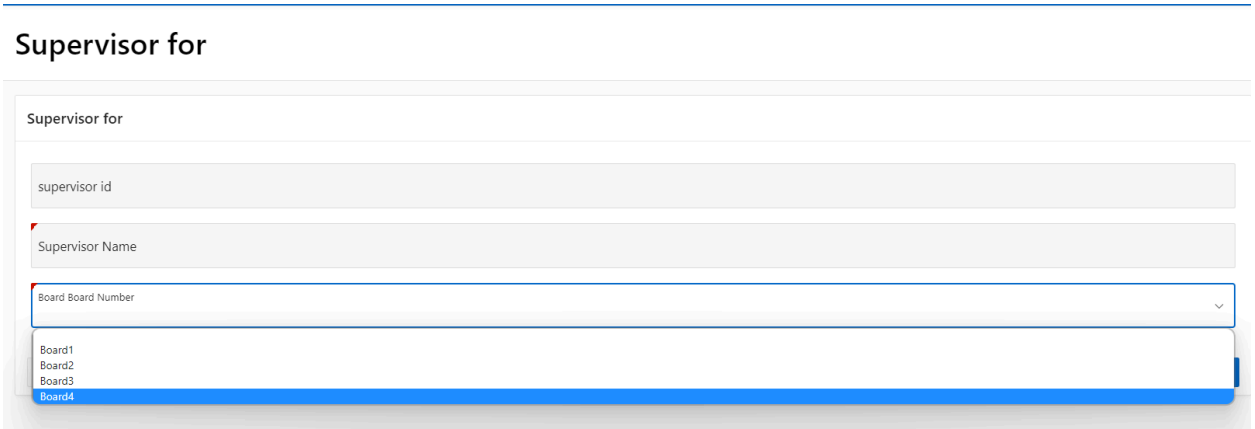

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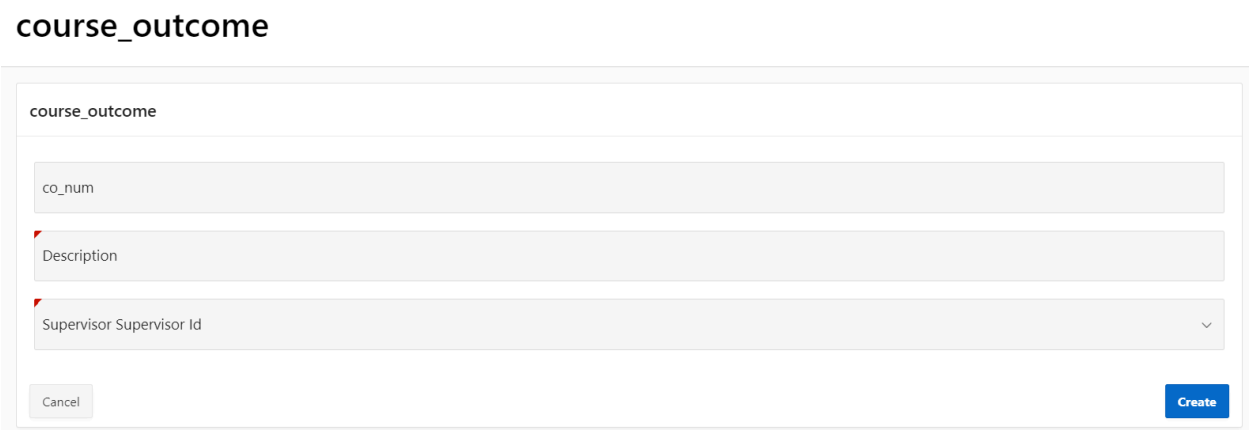

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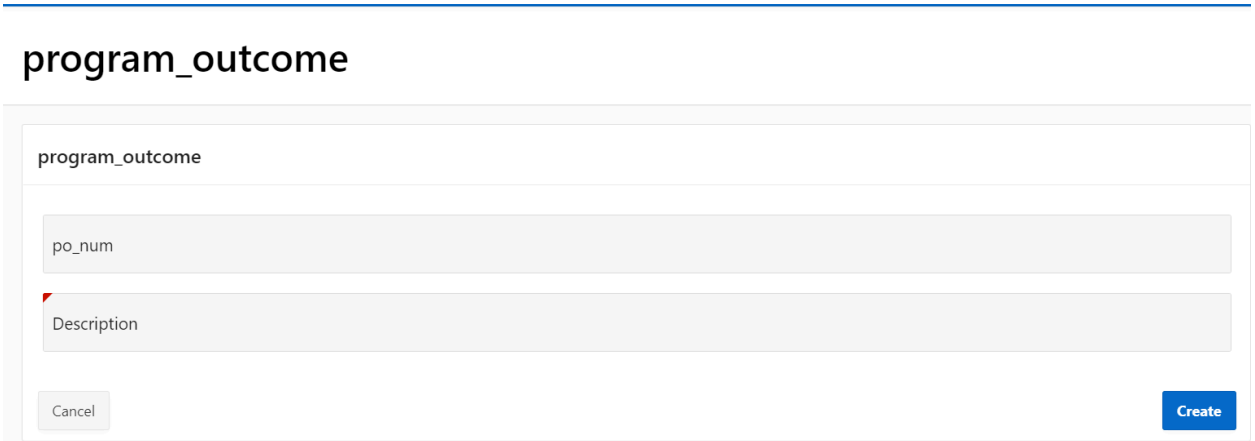
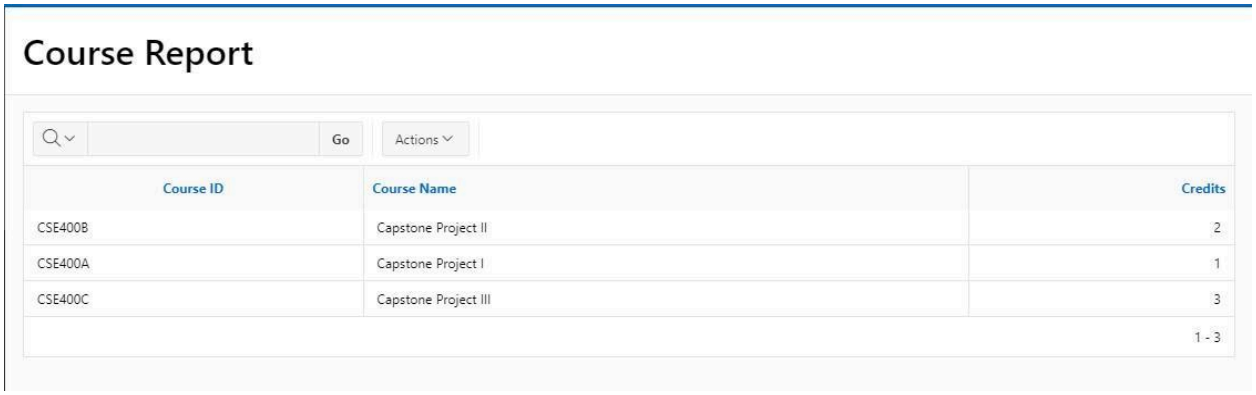


Figure:4.18

Figure:4.19

Figure:4.20



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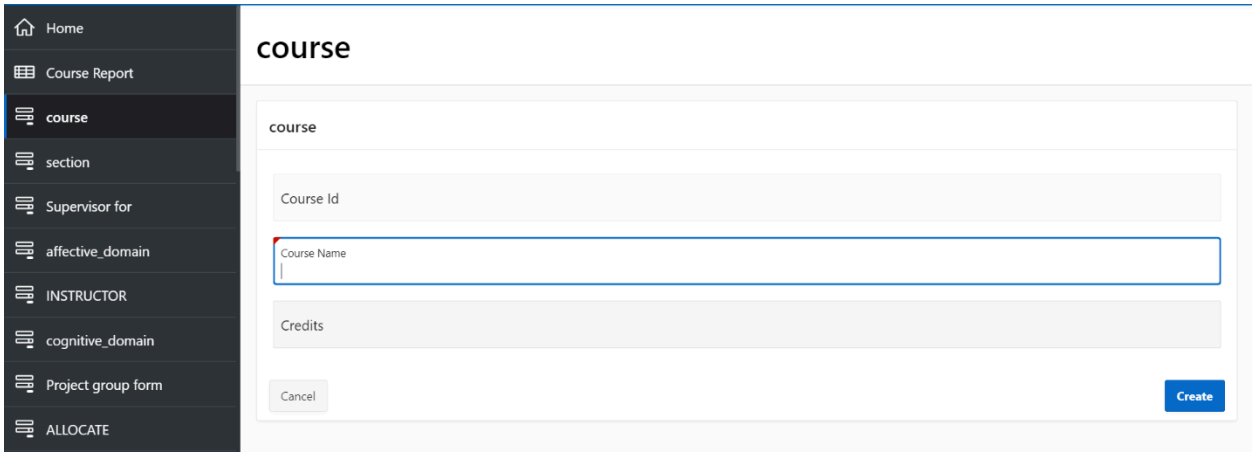

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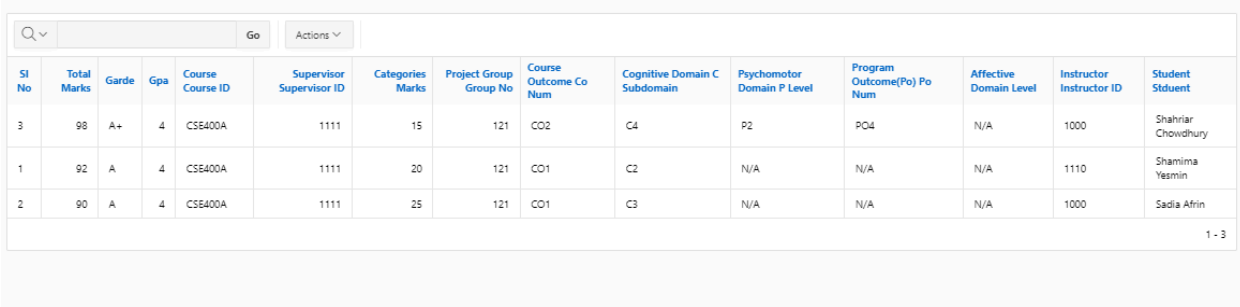

Go | Actions

Sl No	Total Marks	Garde	Gpa	Course Course ID	Supervisor Supervisor ID	Categories Marks	Project Group Group No	Course Outcome Co Num	Cognitive Domain C Subdomain	Psychomotor Domain P Level	Program Outcome(Po) Po Num	Affective Domain Level	Instructor Instructor ID	Student Stduent
3	98	A+	4	CSE400A	1111	15	121	CO2	C4	P2	PO4	N/A	1000	Shahriar Chowdhury
1	92	A	4	CSE400A	1111	20	121	CO1	C2	N/A	N/A	N/A	1110	Shamima Yesmin
2	90	A	4	CSE400A	1111	25	121	CO1	C3	N/A	N/A	N/A	1000	Sadia Afrin

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Figure: 4.23

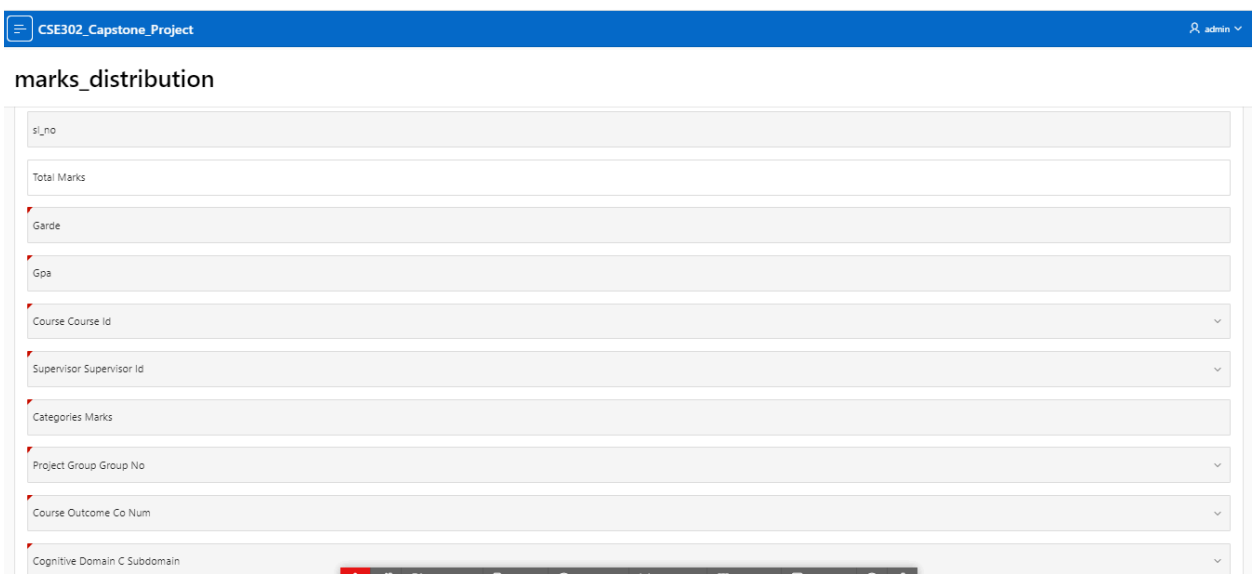

Figure:4.24

Figure: 4.25

5. Master-Detail Form

Page Number and Name	Master Table (Parent)	Detail Table (Child)
92 and course_section_details	course	section
95 and board_instructor_details_	board	instructor
99 and supervisor_co_details	supervisor	Course_outcome

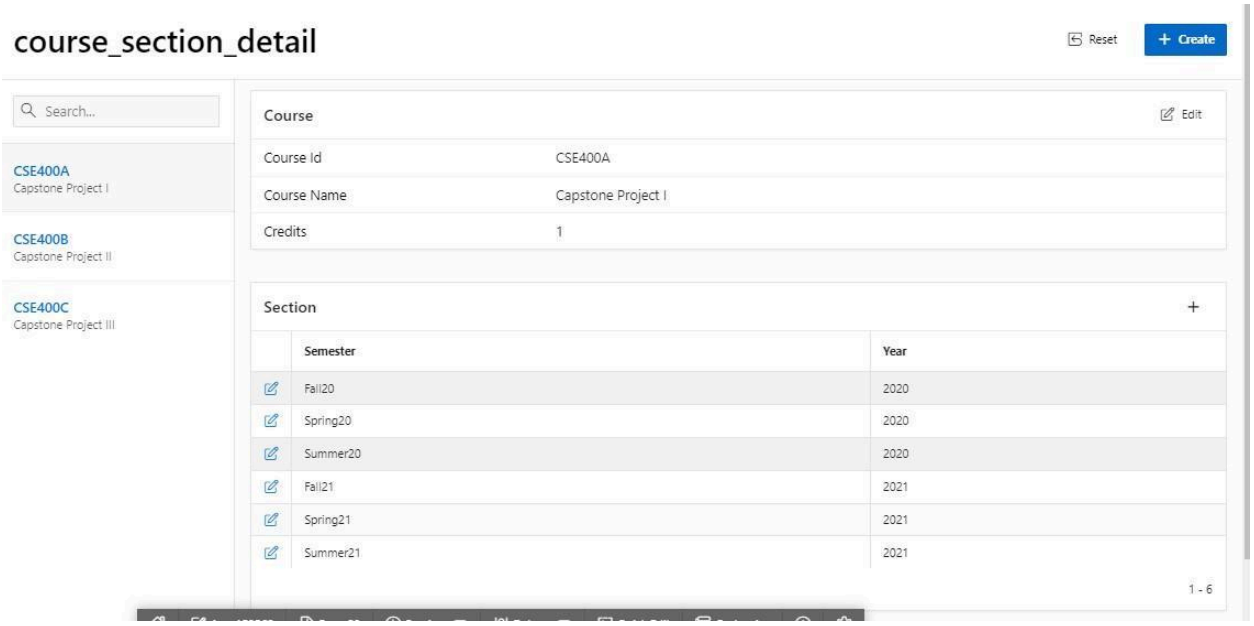

Figure:5.1

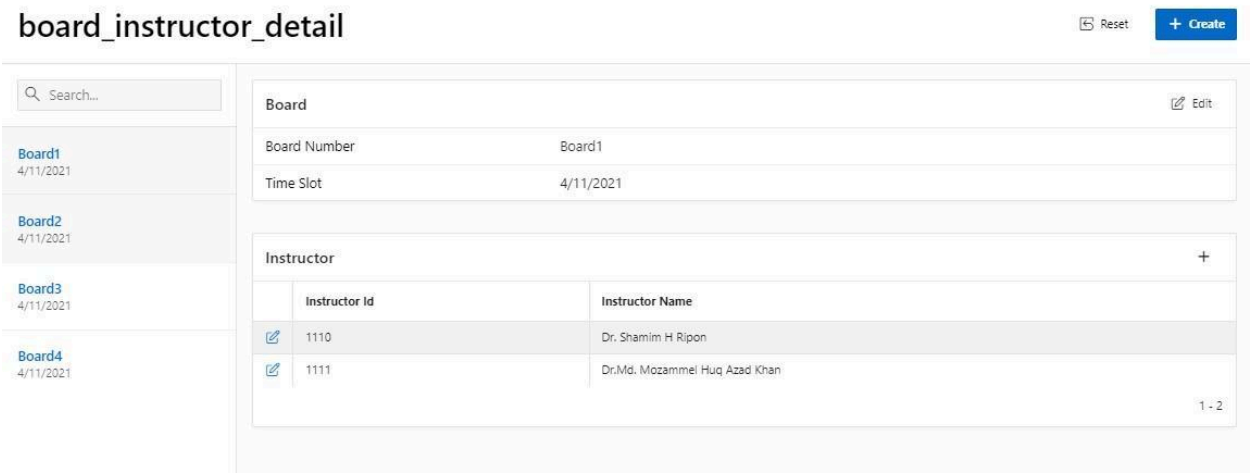

Figure:5.2

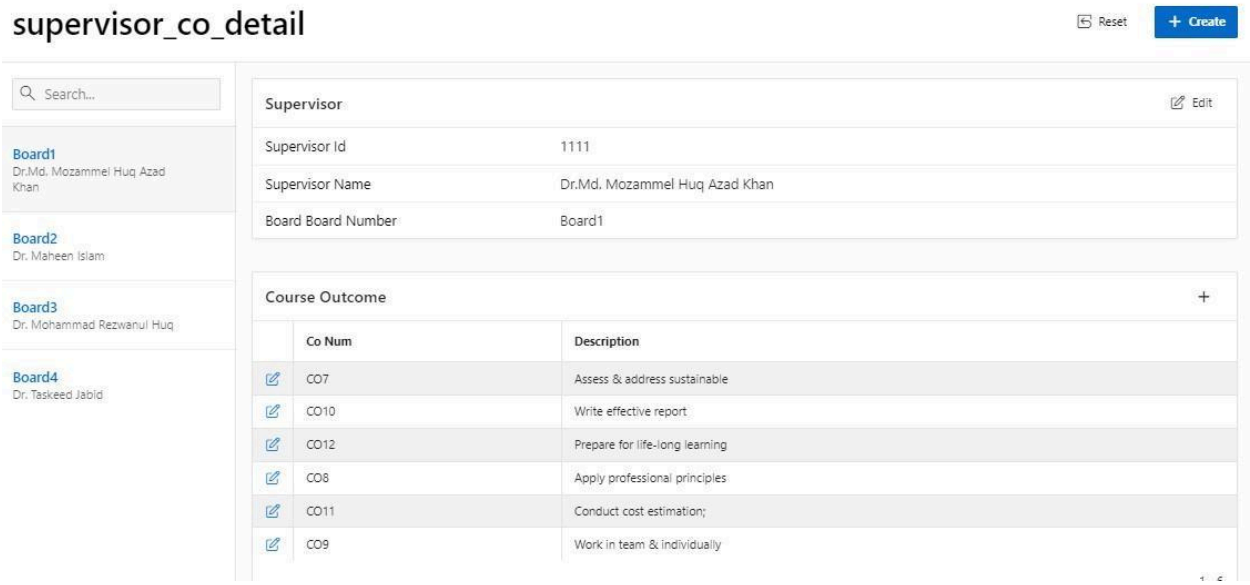

Figure:5.3

6. Reports based on Multi-Table and Aggregate Queries

Page Number and Name	Query Type	Report description

7. Parameterized Reports

Page Number and Name	Given Input	Report description

8. Charts and Other Pages

Page Number and Name	Chart description

9. Authorization and User Role Assignment

User Role Assignment

User name	Password	Role
admin	admin123	Administrator
student	student123	reader
instructor	instructor123	Administrator, Contributor

Page-wise Authorization Information

Page number and name	Authorization Level
P5 and course_report	Reader
P45 and student_report	Reader
P47 and section_report	Reader
P55 and instructor_report	Reader
P40 and student_form	Administrator
P11 and course_form	Administrator
P103 and instructor_form	Administrator
P20 and project_group_no	Administrator
P38 and marks_distribution	Administrator

10. Concluding Remarks

While doing this project, we had to face a lot of challenges. Especially while making the E-R diagram we had to rely on so many assumptions which enforced us to think out of the box. While using the oracle apex, we have spent sleepless nights to complete the project but unfortunately we could not apply the triggering method.